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Geothermal energy resources in Poland – Overview of the current state of knowledge

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ABSTRACT

Geothermal energy, which is one of renewable energy sources, is the internal heat of the Earth. Currently in Poland hydrogeothermal resources are utilized, for which the energy carrier is hot groundwater produced with the wells. Petrogeothermal energy resources, i.e. resources of heat accumulated in rocks for which the energy carriers are media injected through wells into the hot rock formations has not yet been utilized in Poland. However the research work tending to assess the possibility of utilization of this type of energy was carried out. The results are particularly important in the context of forecasts of geothermal energy sector development. Although technology of petrogeothermal resources utilization is in the experimental stage, is considered to be the technology of future. In Poland, low-temperature geothermal resources occurs, related mainly to sedimentary rocks – sandstones, limestones, dolomites, rarely with igneous rocks (crystalline, volcanic). Depending on the hydrogeothermal parameters these resources may be used for different purposes: first of all for heating, but also for therapeutic and recreational purposes. Research of the possibilities of using geothermal resources (both petro and hydrogeothermal) for electricity production are carried out. The article presents the actual state of knowledge about geothermal resources in Poland as well as possibility of them utilization for different purposes. In many parts of the Poland significant unused potential of geothermal waters and energy exist.

1. Introduction

Energy from renewable sources in Poland comes mostly from solid biofuels (ca. 76% in 2014), liquid biofuels (9.2%), wind energy (8.2%), biogas (2.6%) and water energy (2.3%). The use of geothermal energy is still at a marginal level. In 2014, the acquisition of geothermal energy accounted for ca. 0.25% of the energy gained from all RES [1]. In the EU-28, geothermal energy accounted for 3.2% of renewables in primary production in 2014 [2]. Geothermal energy development in Poland is insufficient taking into account the resources of geothermal energy, as well as the potential for their use.

Geological conditions of the geothermal waters occurrence are analysed in Poland since 1980-ties. As a result of the research prospective locations of geothermal water and energy utilization for heating, balneotheraphy and recreational purposes were indicated [3–13]. A relatively new, but very important element of the geothermal resources development, is an indication of the possibilities of using the existing geothermal potential for electricity production. There are areas where the use of geothermal resources for heat and electricity production could result in satisfying outcome. The key research in this field were carried out in the years 2010–2014, when two projects devoted to hydrogeothermal as well as petrogeothermal resources (hot dry or wet rocks – HDR/HWR) utilization for heat and electricity production were accomplished.

The research connected with analysis of the possibility of using hot dry or wet rocks for building the closed geothermal systems in Poland was initialized by the project carried out in the years 2010–2013 by a consortium which includes leading scientific institutions in Poland (the Polish Geological Institute – National Research Institute, the AGH University of Science and Technology, the Mineral and Energy Economy Institute of the Polish Academy of Sciences, and the PBG Geophysical Exploration Co. Ltd.). It was the first enterprise of this type, which tends to recognize the potential of hot dry rocks for heat and electricity production. As a result of the project, prospects for building closed geothermal systems – Enhanced Geothermal Systems (EGS) in the territory of Poland have been indicated. As one of the consortium members, a research team from the AGH University of Science and Technology conducted the work connected with analysis of the possibility of using reservoirs in sedimentary rocks for EGS. The results of the project were presented in numerous publications [14–21].

In 2014 *Atlas of the possible use of geothermal waters for combined production of electricity and heat using binary systems in Poland* [22] was

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issued. The work has been carried out by a scientific consortium of the Mineral and Energy Economy Research Institute of the Polish Academy of Sciences in Krakow, AGH University of Science and Technology in Krakow and the Carpathian Division of the Polish Geological Institute – National Research Institute in Krakow. The main goal of the study was to indicate the suitability of the hydrogeothermal conditions existing in Poland for generating electricity from geothermal waters, as well as to select the most effective method for implementing binary technologies taking account of the existing geothermal potential and local infrastructural conditions.

All the above mentioned works are an excellent source of information about geothermal water and/or energy resources in Poland.

The purpose of this paper is to provide a general overview of geothermal energy resources in Poland (both hydro and petrogeothermal) as well as possibilities of them utilization for energetic purposes – with emphasis on the indication of the possibilities of electricity production in selected areas. Earlier publications on geothermal resources in Poland devoted mostly to hydrogeothermal resource utilization [e.g. [7–10]]. Simultaneously, information about petrogeothermal research in Poland [e.g. [14–16]] not refer to the existing hydrogeothermal resources. Integration of research results provides a coherent information about Polish geothermal potential. The paper shows in particular results of geothermal projects carried out by AGH-UST as well as overview of Polish geothermal resources – both conventional and unconventional.

Geothermal resources in Poland are relatively well-recognized, but them development is very slow. The big problem so far was the lack of financial support. Recently, the financial support of geothermal investments in Poland is provided by the Central Fund for Water Management and Environment Protection. In 2016, 30 applications for co-financing of geothermal investments were submitted. The results of these actions should be visible in the coming years.

2. Possibilities of geothermal energy utilization

Geothermal technologies use renewable energy resources to generate electricity and/or heating and cooling while producing very low levels of greenhouse-gas emissions. They thus have an important role to play in realizing targets in energy security, economic development and mitigating climate change [23]. Geothermal energy is considered renewable as there is a constant terrestrial heat flow to the surface, then to the atmosphere from the immense heat stored within the Earth. Heat can be extracted at different rates. Sustainable use of geothermal energy implies that the heat removed from the resource is replaced on a similar time scale [24]. There are three primary ways we can use geothermal energy: for electricity production, for direct-use applications, and for heating and cooling buildings with geothermal heat pumps [25].

The most favorable conditions for industrial utilization of geothermal energy are connected with high-temperature geothermal systems. This systems occur within the crustal plates, in rifts and in subduction zones. In such zones reserves of water/steam with temperatures over 150 °C occur at the depths from several hundred to several thousands of meters. Their manifestations at the surface are hot springs and geysers. These kind of resources are suitable for power generation as well as direct applications, which include wide range of temperatures and installations. But more common and cover much larger areas in comparison to this high-temperature system are low-temperature systems. Low-temperature systems occurs in most European countries, also in Poland [5]. Key sector for low- temperature systems development is heat generation. But this kind of geothermal resources could be also utilized in many different ways (Fig. 1), for example for therapeutic and recreational purposes. It also could be used to produce power by using technologies such as the Organic Rankine Cycle (ORC) and Kalina Cycle (KC) [26]. Utilization methods of geothermal energy depending on the temperature of the water or steam is presented on Lindal diagram [27] (Fig. 1). The direct use of geothermal energy is the most common use of

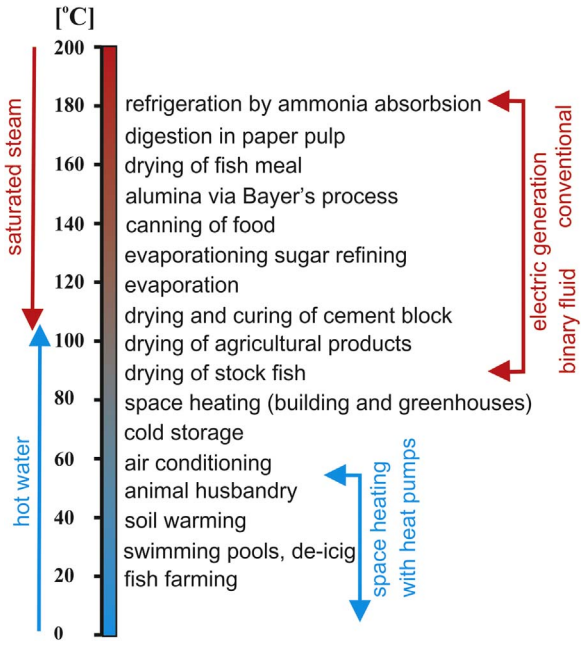


Fig. 1. Lindal diagram [27].

this form of energy, as opposed to its conversion to electricity [28]. Geothermal energy is being used for heating purposes to an ever greater extent. Current trending direct uses are mainly for heating systems working directly or through heat pumps, aquaculture, drying crops, growing plants and vegetables in greenhouses, processes of the paper and the cement industry, food processing, brewing, dyeing of fabrics, snow melting, cooling spaces and balneology, among others [29]. In many cases, using cooled waters for drinking purposes may be considered an alternative method of disposing of them. In some case the technology for desalinating spent geothermal waters with high mineral content need to be used. Geothermal waters are a source of clean energy as well as a valuable product with therapeutic, balneotherapeutic and recreational properties and should be utilized efficiently or utilized in a sustainable manner. Geothermal sources allow for efficient resource utilization due to their high capacity factor [30–34].

Geothermal energy has experienced a renaissance in the past ten years as many new technologies and countries have joined the industry. The technology for generating electricity and deploying district heating from high-grade hydrothermal systems is relatively mature and reliable. Technologies for geothermal heat pumps are also mature and are being deployed at increasing rates in the United States and Europe. The use of innovative hybrid and combined heat and power plants, lower resource temperatures, and enhanced reservoir stimulation methods are making geothermal energy accessible in a much greater variety of places [35].

3. Geothermal resources in Poland

The geothermal energy resources are defined as total amount of thermal energy (heat) accumulated in the Earth's crust down to given depth, referred to particular area for which the calculations are made and for mean annual temperature at the Earth's surface groups. Conventionally, geothermal resources are hydrothermal resources that include reservoirs of hot water and/or steam, and are categorised as vapor-dominated or liquid-dominated resources. Hydrothermal resources exist at shallow to moderate depths and are the least abundant source of geothermal resource. Other geothermal resources (unconventional) include geo-pressured, magma and the more widespread hot, dry rock (petrogeothermal energy) [36].

In contrast to hydrothermal resources, petrogeothermal resources are connect not with water and/or steam but with hot rocks. This kind

of reservoirs are, instead, man-made reservoirs in rocks that are artificially fractured, and thus any convenient volume of hot dry rock in the earth's crust, at accessible depth, can become an artificial reservoir [25]. The energy carriers are media (usually water) injected through wells into the hot rock formations. Petrogeothermal resources are the subject of global research on the technology of heat recovery for power generation in binary systems, often in combination with heat production. The history of their use is much shorter than conventional resources. In 1970 (Los Alamos, USA) HDR (Hot Dry Rock) system was proposed as a method for exploiting the heat contained in those vast regions that contain no fluids in place (petrogeothermal resources). HDR system recovers the Earth's heat via closed-loop circulation of fluid from the surface through a man-made, confined reservoir [37]. Later works reveal that in some cases reservoir rocks contain a small amount of groundwaters. The utilization system of this kind of petrogeothermal resources was called EGS (Enhanced or Engineering Geothermal System). EGS was defined as engineered reservoirs that have been created to extract economical amounts of heat from low permeability and/or porosity geothermal resources. EGS include conduction-dominated, low permeability resources in sedimentary formations [38]. In EGS, the naturally occurring hot rock does not contain enough water and generally lies at greater depth than is typical of hydrothermal systems. Enhanced or engineered geothermal systems have increased the number of locations that could be suitable for geothermal power production [39].

Currently in Poland low-temperatures hydrogeothermal resources are utilized. The majority of Polish geothermal resources are low exergy resources, with the exception of some places where medium exergy resources exist (Konin area) [40].

For several years the research connected with assessment of analysis of the possibility of using petrogeothermal energy are carried out. The results of the work show that there are both hydrogeothermal as well as petrogeothermal resources.

For all kind of geothermal systems, flow rate Q and the temperature of the reservoir fluid T are the key parameters that control the power output P of a geothermal power plant. For large-scale resource assessments the temperature in the subsurface is a relatively convenient parameter to work with. In most nonvolcanic areas in Europe, conduction is the dominant heat transfer mechanism in the lithosphere. Temperatures can therefore be estimated with a steady-state conductive model based on assumptions and inferences on the thermal conductivity structure of the lithosphere, the heat flow at the base of the lithosphere, and on the content of heat-producing elements in the lithosphere [41]. Flow rate depends strongly on the (enhanced) reservoir permeability, determined by lithological properties such as porosity, and the presence, distribution and permeability of natural fractures. These properties are dynamic and will change when the area of the reservoir is subjected to changes in temperature, pressure and the state of stress. Therefore, the reservoir permeability can easily vary by several orders of magnitude [42].

3.1. Hydrogeothermal energy resources

Hydrogeothermal energy resources in Poland are accumulated in underground waters reservoirs in various stratigraphic units and at various depths in four main hydrogeothermal provinces.

The largest in area (approximately 87% of the country) and the most perspective is a province of the Polish Lowlands (Fig. 2). Particularly favorable conditions for the utilization of geothermal energy are in the region of the Polish Trough, where geothermal energy is currently used in several locations (for heating, bathing and balneotheraphy), and several other projects are in progress at various stages of advancement. In many areas of the Polish Lowlands utilization of geothermal waters with relatively high temperatures (even in excess of 100 °C) and high capacities (even 300 m^3/h) is real [43,44]. Problematic issue in some regions could be corrosion of geothermal installation connecting with high value of TDS, sometimes even 300 g/dm^3 (especially in zone of salt diapirs occurrence). Reservoirs are built mostly from sandstones, locally also carbonates. The main resources of geothermal waters in the Polish Lowlands are reservoired in the Mesozoic groundwater horizons. Paleozoic aquifers are characterized by lower values of resources [45–47]. Geothermal waters are accumulated first of all in the Lower Jurassic and Lower Cretaceous formations but significant resources of geothermal energy are reservoired also in the Upper Jurassic, Middle Jurassic, Upper Triassic and Lower Triassic formations. The most interesting and promising areas occur in the Warsaw Trough, Mogilno-Łódź Trough (in the central part of Poland) and Szczecin Trough (in the northwestern part of the Polish Lowlands). Utilization of thermal waters for heating purposes in particular voivodeships and towns of central Poland should, first of all, be based on the resources of the Lower Jurassic aquifer [7].

The temperature field of the Lower Jurassic formation in the Polish Lowlands changes significantly, from those typical of subsurface waters (outcrops, zones of direct recharge) to 120 °C in the deep, axial part of the Łódź Trough (area north and northeast from Konin town). Temperatures over 50 °C occur in the Warsaw, Łódź and Szczecin troughs. In most part of Lower Jurassic aquifer potential discharge of wells exceeds 100 m^3/h. Highest discharges (about 300–450 m^3/h) are predicted for northeastern part of the Łódź Trough. In the Szczecin, Warsaw and remaining part of the Łódź troughs as well as in the Kujawy Block discharges about 250–350 m^3/h can be presumed. The TDS of Lower Jurassic groundwaters is in the range of few to over 200 g/dm^3 in the axial parts of troughs. The predicted thermal power of geothermal dublets is up to 30–50 MW_{th} [5].

The temperature of the Lower Cretaceous aquifer is closely linked to the depth of occurrence. Only in some areas: the Skierniewice-Płock region, the northwestern part of the Łódź Trough (Konin vicinity), and the northwestern part of the Szczecin Trough temperatures rise to over 50 °C being related to the deepest structural depressions along the central parts of these units. Highest temperatures were observed in the Konin area (over 90 °C). Therefore perspective area of Lower Cretaceous aquifer occurrence is more limited. However, in these area high discharges of potential wells (over 100 m^3/h, locally even over 300 m^3/h) are predicted. The TDS values do not exceed 40 g/dm^3. The thermal power values of hypothetical geothermal dublet were calculated for the range of 2–20 MW in central part of Polish Lowlands [5].

The next prospective region for hydrogeothermal energy development in Poland is the Podhale area, located in the Carpathian Geothermal Province (Fig. 2). In the Polish part of the Carpathians, the best reservoir and exploitation properties for geothermal water utilization occur just in Podhale. The most prospective aquifer (subject of exploitation) occurs within the Middle Triassic limestones and dolomites and in overlying Middle Eocene carbonates at the depths of 1–3.7 km. These formations are found over the entire Podhale system, prolonging to the Slovakian territory. The maximum reservoir temperatures within the main aquifer (depths of 2–3.2 km) are as high as 90–95 °C. Within the deeper parts of the system, the measured formation temperatures reach 120–128 °C (depths of 4.5–4.8 km). The artesian flow rates from individual wells vary from several to 550 m^3/h. The TDS of geothermal waters are at the level of 0.1–3.2 g/dm^3 [48]. Podhale is a region in the Western Carpathians where geothermal waters have been recently utilized and will be utilized in the future, preferably for heat generation but also for recreation and balneotherapy. The Podhale heating system is currently the largest geothermal installation in Poland. The total capacity of the geothermal heating system is ca. 40.8 MW_{th}, while the total installed capacity of the heating plant is 80.5 MW_{th} [49]. Possibilities of geothermal energy utilization in remaining areas of Carpathian are rather low and related to limited areas [8].

Less perspective are related to Fore-Carpathian Geothermal Province. The province is located in the southern part of Poland covering the area of the Carpathian Foredeep (Fig. 2). In this area

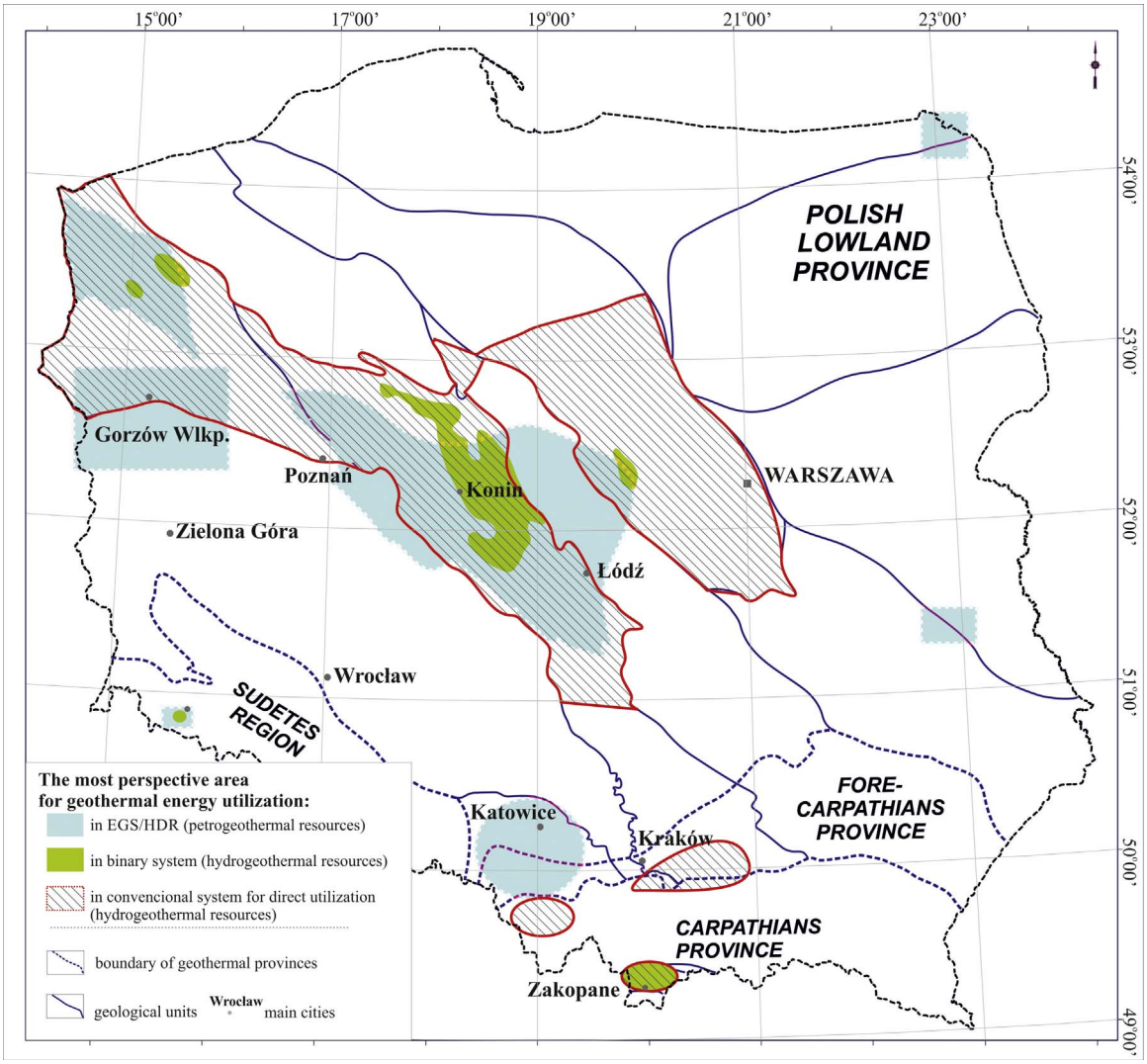


Fig. 2. Location of the most prospective area for geothermal energy utilization in Poland (based on [4]).

geothermal potential related to utilization of geothermal waters of the Carpathian Foredeep for recreation and/or balneotherapy is much higher than that related to utilization for heating purposes. It results as well from the lower energy demand of geothermal water intakes as from favorable physicochemical parameters of these waters, confirmed by numerous wells drilled over vast areas of the Carpathian Foredeep. In the Carpathian Foredeep, low discharges of wells in almost all the analysed aquifers present the fundamental problem (mostly below 20 m^3/h). The Cenomanian aquifer is an exception, as high discharges can be expected over the whole area of its occurrence (up to 250 m^3/h). Temperatures of groundwaters in Cenomanian reservoirs vary in wide range from 30 to 100 °C. The best reservoirs parameters of the Cenomanian sandstones occurs in the central part of the Carpathian Foredeep (Fig. 2). Zones with increased potential discharges of wells are sporadically encountered in the remaining reservoirs. Favorable parameters for heating purposes are locally revealed by waters accumulated in: the Devonian and Carboniferous carbonate sequences of the western part of Carpathian Foredeep, the Middle Jurassic sandstones in the eastern part of analysed area and the Upper Jurassic carbonate rocks in the area to the south of Brzesko, under the Carpathian overthrust (Fig. 2) [50].

Definitely different geothermal conditions are present in the Sudetic Geothermal Region (SW Poland) (Fig. 2). Geothermal waters occur in this region only in the crystalline formations of the Sudetes. Most of the fragmentary hydrogeothermal investigations carried out so far in the Polish part of the Sudetes were limited to zones of occurrence of thermal waters utilized for therapeutic purposes, or to a few areas in which prospection has been carried out for such waters [51]. Currently, the waters of this region are used in several towns for balneotherapy. Sudetic Geothermal Region is characterized by favorable thermal conditions (Cieplice) what it is undoubtedly a great asset of this region.

In the areas characterized by the best hydrogeothermal conditions main structures prospects for use of binary technologies were indicated. The most prospective region for hydrogeothermal energy utilization (including electricity production) is Podhale Basin. In the Polish Lowlands the best prospects for the utilization of geothermal water in binary systems occur in the central part of Polish Lowlands, especially in the Konin area where the temperature of the water accumulated in Lower Jurassic reservoir exceed 90 °C, as well as the Szczecin Through area. Due to existence of geothermal anomalies Sudetic Region appears to be appropriate for the location of binary system. In Cieplice water with temperature 86.7 °C was obtained from the depth of 2002.5 m [51].

3.2. Petrogeothermal energy resources

Petrogeothermal energy resource differ significantly from conventional geothermal reservoirs.

This kind of resources in Poland have been recognized by scientifics teams conducting research in three different geological conditions –



Fig. 3. Location of geothermal installations in Poland (based on [60]).

sedimentary, volcanic and crystalline rocks. Critical requirements for the EGS or/and HDR location in all geological conditions comprise [37,38,52,53] thermal parameters of the rocks (temperatures > 150 °C), thickness of the reservoir (minimum 300 m), porosity and permeability of the reservoir rocks (as the lowest) and the depth of the reservoir (3–6 km).

The performed analyses for volcanic rocks allowed to determine two prospective locations (Fig. 2) - Gorzow block area (Permian volcanics) and the Parczew area (Visean and Ediacaran volcanism). Gorzow block was selected due to the geological analogy to volcanic reservoir located in Gross Schoenebeck near Berlin, where the EGS project is implemented. In Parczew area Proterozoic basalts are most prospective. Dębno area located on the Gorzow block was selected as the optimal location for the HDR/EGS in volcanic rocks. These region is characterized by favorable thermal and geological conditions. Geothermal gradient, amounting in this region approx. 40 °C/km, implies a high temperature of rocks at a relatively shallow depth, allowing production of electricity in binary systems. The temperature within the reservoir at a depth of 4.3 km is ca. 153 °C [14].

Most promising area for exploration of geological structures suitable for the deployment of hot dry rock technology within crystalline rocks was identified in Karkonosze pluton – particularly in the south of Szklarska Poręba and south-east of Jelenia Gora. Geothermal gradient in these zones amounts to approx. 4 °C/100 m. At a depth of 4000 m bsl rock temperature reaches approx. 165 °C [14].

The work carried out for the sedimentary cover allowed to indicate three prospective areas. In the Polish Lowlands, two areas become apparent (Fig. 2). The first area encompasses partly the region of the Szczecin Trough and the north-western part of the Fore-Sudetic region, whereas the second area is situated in the region of the Mogilno-Łódź Trough and fragmentarily in the Kujawy Swell region. In the Polish Lowlands, rock formations lose their reservoir properties at depths deeper than 3–3.5 km and reservoir intervals at great depths are closely connected with fractures and joints, not with intergranular porosity. As a result of the analytical work, an area situated in the central part of Poland was chosen as prospective for EGS in sedimentary rocks in Poland. In the Carpathians and Carpathian Foredeep (south part of Poland), the central part of the Upper Silesian Block may be considered to be prospective. Upper Silesia is conspicuous by a zone of high heat flow values. Heat flow values rapidly increase in foldbelt areas of the Variscan externides and in the northern zone of the Lower Silesian internides, reaching the maximum level in Poland of over 107 mW/m^2 [54]. However, finally it was decided that the most promising conditions for the Enhanced Geothermal Systems occur in the central part of Poland in an area covering the Mogilno-Łódź Trough region and a small part of the Kujawy Swell and Fore-Sudetic regions [55]. In the selected area prospects for EGS have been indicated in the Middle Triassic, Lower Triassic, Lower Permian and Carboniferous rocks [17]. There are several potential places for location of closed geothermal systems. Buntsandstein strata, which seem to be a complex meeting the EGS requirements in the investigated area very well. The most prospective area for the location of Enhanced Geothermal System in sedimentary rocks in Poland is situated in the vicinity of Krośniewice/Kutno, where the top of the Lower Triassic reservoir of more than 1000 m thick. The temperature in the depth of 5.5 km exceed 179 °C. The porosity of reservoir rocks is approximately 3% and the permeability is in the range 0.02–0.1 mD [18].

4. Current utilization of geothermal resources in Poland

Geothermal waters, vapor and energy have thousands-years-long history of utilization [56]. One can find these features in myths, legends, religious ceremonies as well as in balneology and medicine, social and cultural life, literature, history and environment protection.

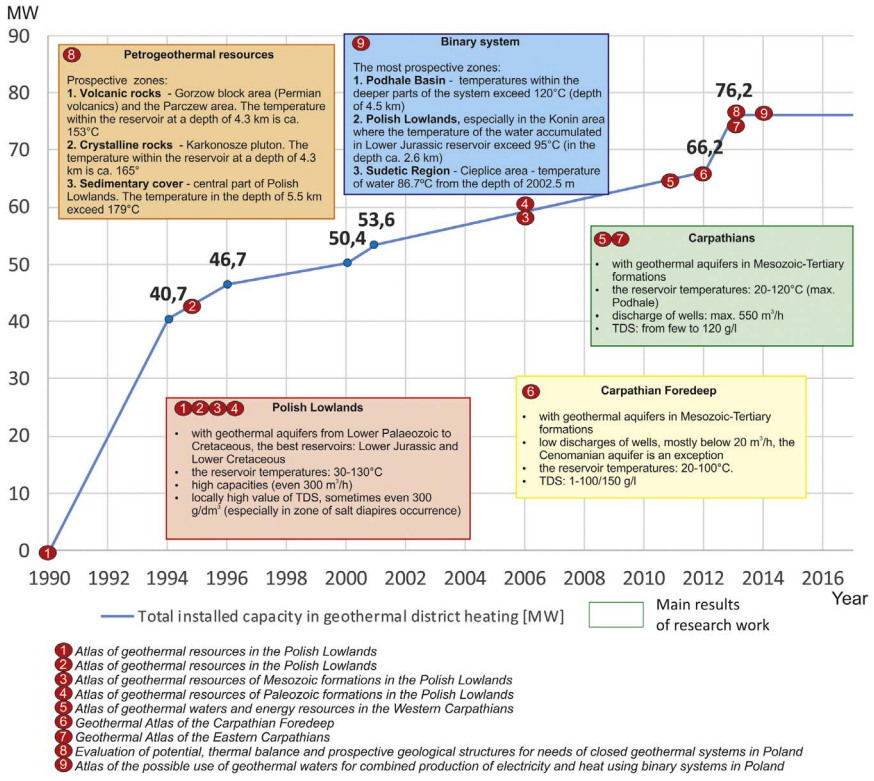


Fig. 4. Total installed capacity in geothermal district heating in Poland on the background of the research work results.

Also in Poland the warm spring in Jaszczurowka near Zakopane was mentioned in folk tales told by Sabała - a XIX-century story-teller, musician and robber from the Tatra Mts [57]. Natural thermal springs were known from time immemorial in the Polish part of the Sudetes at Cieplice and Lądek [51].

Currently in Poland low-temperature hydrogeothermal resources are used. A key sector for their development is space heating. The activities on geothermal heating deployment were initiated in Poland in the 1980s. Since that time six district heating plants have been launched (Figs. 3 and 4) in the Podhale region (since 1994), in Pyrzyce (since 1996), in Mszczonów (since 1999), in Uniejów (since 2001), in Stargard (since 2012, re-opened after closure in 2008) and in Poddębice (since 2013). In addition, individual heating systems were in some recreation centers. In 2015 their total installed geothermal capacity was ca. 76.2 MWth and heat sales ca. 817 TJ. Most geothermal heat was produced by the Podhale plant: ca. 462 TJ/2015 – one of the largest geothermal district heating plants in continental Europe [58].

In recent years a growing interest has been observed in the recreation and balneotherapy sector. In 2015 thirteen spas (with formal status of health resorts) using geothermal water for curing were operating. One may estimate their total capacities for at least 20 MWth and heat uses for at least 210 TJ in 2015. Other minor uses comprised semi-technical wood drying (some 0.2 MWth, 0.6 TJ and heating up of football playground and walking path (1 MWth, 5 TJ) as well as aquaculture (0.6 MWth, 10 TJ). In 2013–2015 eight new geothermal wells were drilled (one was negative). They encountered ca. 30–95 °C waters for bathing, recreation and space heating in some cases. One injection well was deepened [59].

Total installed capacity in geothermal direct use in Poland in 2015 (geothermal district heating and other geothermal direct use) was estimated ca. 105 MW_{th}. Unfortunately, compare to past years, rapid increase of installed capacity has not been observed. In 2012 total installed capacity was estimated ca. 101 MW_{th}, and predicted installed capacity for 2015 was ca. 130 MW_{th} [60]. Compared to other European country (Fig. 5), development of geothermal energy in Poland is very slow. The Netherlands is the top in relative growth (65% per year),

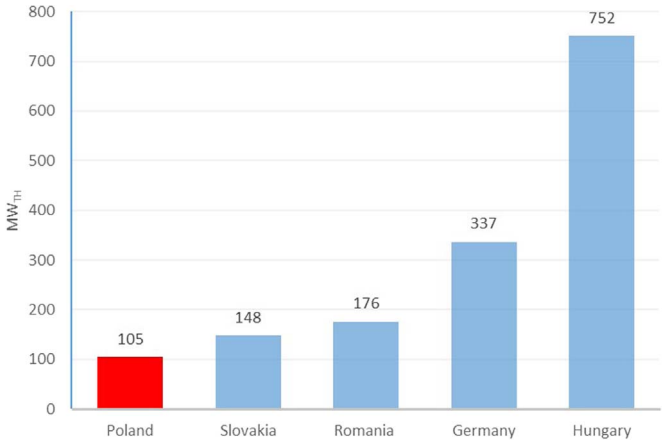


Fig. 5. Total installed geothermal capacity in selected countries of Central and Eastern Europe.

Iceland is champion in geothermal coverage of heat demand (66%) and Turkey is the leader in total amount (2844 MW_{th}). Some have a good development in geothermal direct use, e.g. Slovakia (148 MW_{th}), Romania (176 MW_{th}), Germany (337 MW_{th}), Hungary (752 MW_{th}) [61]. Hungary is one of five worldwide leaders with the largest annual energy use (9573 TJ) [62].

Total installed capacity of geothermal heating from medium to low temperature sources in Europe exceeds 9200 MWth, of which about half is used in district heating [61]. Share of Polish geothermal installed capacity in this amount is about 1.1%.

Worldwide leaders in direct utilization of geothermal energy without geothermal heat pumps are: China (6089 MW_{th}), Turkey (2844 MW_{th}), Japan (2086 MW_{th}) and Iceland (2035 MW_{th}) [62].

5. Summary

Geothermal energy still is a marginal contributor to the Polish

energy mix. As one of renewable energy sources, this kind of energy can be important for numerous regions. Development of geothermal energy gives people the potential to gain better control of their own local energy resources and use a secure, environmental friendly and domestic source of energy. This internal energy of the Earth can be used for different purposes to improve environmental quality and protect public health and safety. The technological and sustainable development of this type of energy will help to solve the world's energy needs and challenges.

In recent years geothermal energy development in Poland has been slow. Despite favorable geothermal conditions in many regions there are still legal, economic or fiscal barriers to increasing level of geothermal resources utilization. Until now, geothermal energy is not a significant source of heat in Poland and is not a source of electricity at all.

Poland has low-temperature geothermal resources connected mostly with Mesozoic sedimentary formations. At present geothermal applications involve space heating, balneotherapy, bathing and recreation, nevertheless, research on the possibilities of using geothermal energy for other purposes (also electricity production) are carried out. At present (2016) only hydrogeothermal energy (connected with geothermal water) is utilized. Although petrogeothermal energy that constitutes heat resources of rocks has not yet been utilized, the research work tending to assess the possibility of utilization of this type of energy is carried out.

Hydrogeothermal energy resources in Poland are accumulated in underground water reservoirs in various stratigraphic units and at various depths in areas of the Polish Lowlands, the Carpathians and in some locations in the Sudety Mts and Carpathian Foredeep. The most perspective area is connected with the Polish Lowlands where geothermal water resources are reservoirs in Mesozoic aquifers, particularly in Lower Cretaceous and Lower Jurassic clastics rocks. Mogilno-Łódź Through is characterized by the most favorable geothermal conditions. However Warsaw and Szczecin Troughs should be also considered as promising areas for geothermal energy development.

Big prospects should be associated with the further development of EGS technology. The world predictions indicate that the petrogeothermal energy is the energy of future and in the coming years development of such energy sector can be expected. Also in Poland the occurrence of the petrogeothermal energy resources possible to be utilized in future has been observed. Considering the resources accumulated in sedimentary, volcanic and crystalline rocks, it should be noted that the most abundant resources occur in three locations: in the central part of the Polish Lowlands (the Mogilno-Łódź Trough). Gorzów block (Dębno area) and Karkonosze pluton (Szklarska Poręba area).

It is worth emphasizing that significant unused potential of geothermal waters and energy exist in Poland. According to European Geothermal Energy Council [63] geothermal energy will be a key source in the European energy mix. In order to realize its full potential to the benefit of European economies and citizens alike, it needs increased and dedicated support now. It also refers to Poland, where low-temperatures geothermal resources development with appropriate support system could provide quantifiable social and economic benefits.

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